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# Surveillance and GIS for Public-Health Decisions

From reported events to a defensible spatial recommendation

Where should a health office verify, investigate, allocate resources, and communicate?

20-30 minute teaching module | Evidence baseline checked 10 August 2026

## What you will be able to do

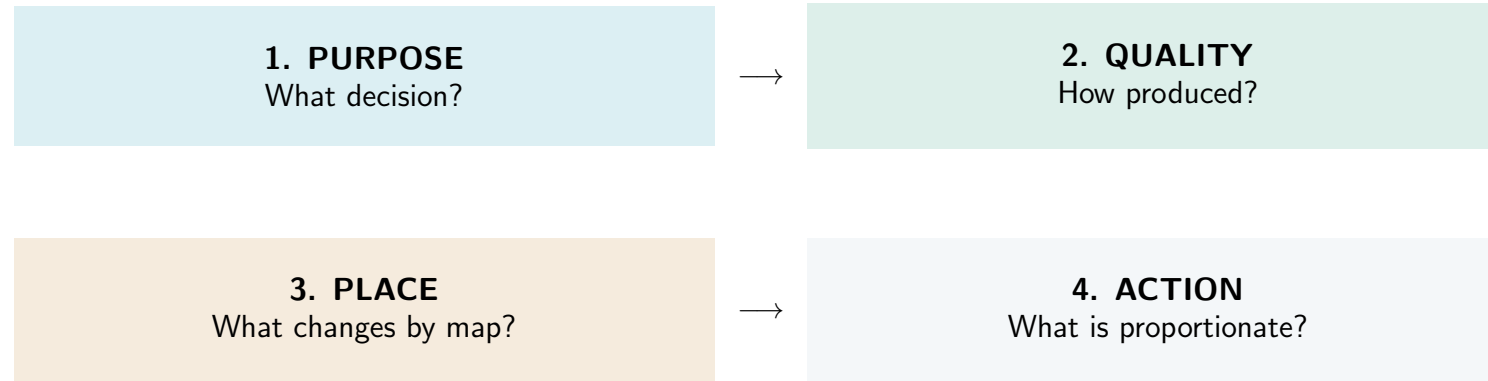
Distinguish surveillance  
from research by purpose

Evaluate data quality  
and system performance

Compare counts, rates,  
and geographic choices

Recommend action  
without causal overreach

## Roadmap: four moves from data to decision



## The case hook

A fictional regional health office receives four weeks of respiratory-event reports from six service areas.

Leadership asks: **Where is the burden? Where is the reporting problem? What can we show publicly?**

Your job is not to diagnose a community from a map. It is to decide what the evidence can support now—and what should happen next.

## First participation prompt

### **OBSERVATION**

What was recorded?

### **INTERPRETATION**

What might it mean?

### **CAUSAL CLAIM**

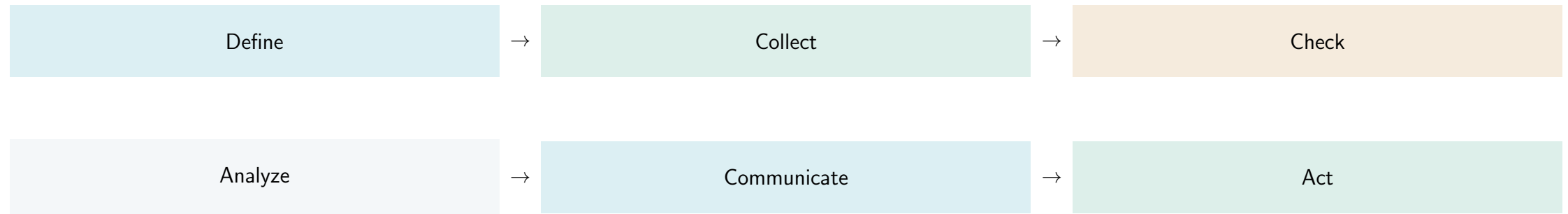
Why did it happen?

### **ACTION**

What should we do next?

Turn to a partner: give one sentence in each category.

## Surveillance is a decision system



Feedback closes the loop: information must return in a useful form.

## Surveillance or research? Start with purpose

### **SURVEILLANCE / PROGRAM**

Detection, control, response, or improvement

### **RESEARCH**

Develop or contribute to generalizable knowledge

The data source alone does not decide. Purpose, design, intended use, and governance matter.

## A count has a production history

|                              |                                                          |
|------------------------------|----------------------------------------------------------|
| <b>Case definition</b>       | Who qualifies as a reported event?                       |
| <b>Reporting opportunity</b> | Who can enter the system—and who cannot?                 |
| <b>Transmission</b>          | What delay, loss, or duplication occurs?                 |
| <b>Analytic rule</b>         | What cleaning, grouping, or threshold changes the total? |

A changing count may reflect changing disease, changing detection, or both.



## Evaluate what matters for this decision

|                    |             |                           |
|--------------------|-------------|---------------------------|
| Simplicity         | Flexibility | Data quality              |
| Acceptability      | Sensitivity | Predictive value positive |
| Representativeness | Timeliness  | Stability                 |

There is no universal passing score. System objectives set priorities.

## Tradeoffs are normal

**FASTER**  
may be less complete

**MORE SPECIFIC**  
may miss events

**MORE DETAIL**  
may reduce acceptability or privacy

## The synthetic six-area dataset

| Area       | Population | Events | Complete | Late |
|------------|------------|--------|----------|------|
| Northgate  | 52,000     | 78     | 94%      | 7%   |
| Riverbend  | 18,500     | 42     | 82%      | 24%  |
| Central    | 81,000     | 96     | 97%      | 4%   |
| Lakeside   | 9,200      | 6      | 76%      | 18%  |
| Hillview   | 31,500     | 60     | 90%      | 9%   |
| Southpoint | 4,800      | < 5    | 88%      | 11%  |

All names and values are synthetic instructional data. They are not official data. < 5 is a classroom suppression rule, not a universal threshold.

## Counts and rates answer different questions

**CENTRAL: 96 events**

Highest workload

**RIVERBEND: 22.7 per 10,000**

Highest displayed rate

Both can be correct priorities—for different decisions.

## Calculate one rate—then audit it

$$\frac{42 \text{ reported events}}{18,500 \text{ people}} \times 10,000 = 22.7$$

Do numerator, denominator, geography, and period describe the same population?

## A signal can also be a data-quality warning

### **RIVERBEND**

22.7 per 10,000

82% complete

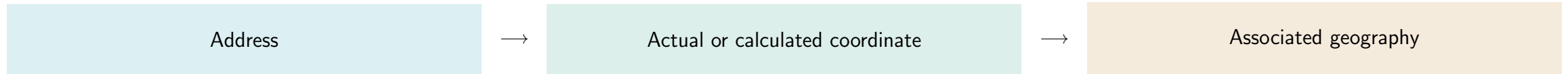
24% late

Before escalation:

- Verify the workflow change
- Reconcile late reports
- Check denominator fit
- Examine persistence over time

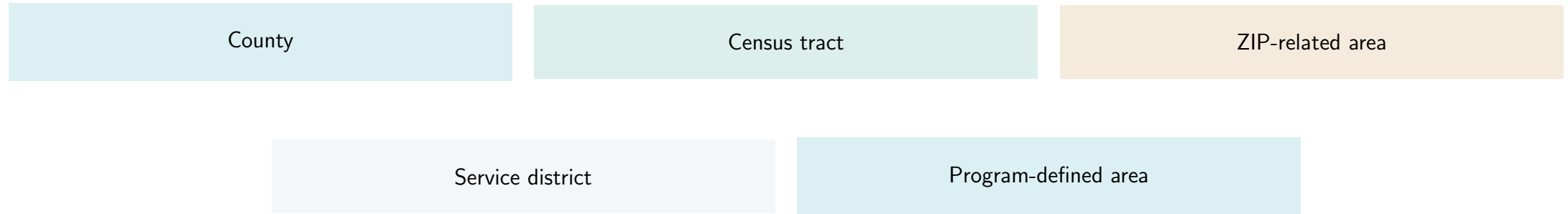
Highest displayed rate does not mean confirmed highest underlying risk.

## Geocoding links records to place



Record match status, approximation, unmatched records, benchmark/vintage, and permitted handling. Address-range coordinates may be interpolated.

## Geographic units are analytic choices



Administrative convenience is not epidemiologic truth.



## A boundary file is not health data

Boundary + vintage

+

Health numerator

+

Population denominator

Units and time must be compatible before mapping.

## The same values can tell different visual stories

### **Raw-count choropleth**

May mostly display where more people live.

### **Rate choropleth**

Improves comparison, but does not cure denominator error or small-number instability.

### **Quantile classes**

Balance areas across colors, but class limits depend on this dataset.

### **Meaningful breaks**

Support a decision only when the thresholds have a defensible basis.

Show the classification rule and underlying values. Color does not create evidence.

## Scale changes both usefulness and risk

### **LARGER AREAS**

More stable  
More private  
Less local detail

### **FINER AREAS**

More local  
Less stable  
Greater disclosure risk

Choose the minimum geographic detail needed for the decision.

**A spatial pattern is a question, not a cause**

**Area-level pattern  $\neq$  individual cause**

A pattern can guide verification, outreach, sampling, or study design.

Avoid ecological fallacy: an area characteristic and area rate do not establish an individual exposure–outcome relationship.

## Privacy changes the analysis

### **SUPPRESSED**

Information withheld

### **ZERO**

Observed value of none

Do not infer, back-calculate, or shade a suppressed value as zero.

Fine geography, rare combinations, and repeated releases can disclose more than any one table.

## Team decision: seven minutes

**ANALYST**  
Calculate and audit

**SKEPTIC**  
Challenge inference

**DECISION LEAD**  
Choose next action

Produce: one immediate action, one verification step, one additional data element, and one privacy-aware map specification.

Be ready to defend why the action is proportionate to the evidence.

## The three-sentence decision brief

**The data show...** observation with count/rate, place, and period.

**The data do not establish...** uncertainty, missingness, and causal limit.

**We recommend next...** proportionate action plus verification.

# Key takeaways

- 01 Start with the public-health decision and purpose.
- 02 Audit how the data were produced before interpreting the pattern.
- 03 Align numerator, denominator, geography, period, and boundary vintage.
- 04 Treat mapping, classification, scale, and suppression as analytic choices.
- 05 Match the strength of action to the strength of evidence.



# Sources and currentness

## Official sources used

- CDC, [Updated Guidelines for Evaluating Public Health Surveillance Systems](#).
- CDC, [Field Epidemiology Manual: Describing Epidemiologic Data](#).
- CDC, [Environmental Public Health Tracking](#).
- CDC, [CDC WONDER Frequently Asked Questions](#).
- U.S. Census Bureau, [TIGER Data Products Guide](#).
- U.S. Census Bureau, [Census Geocoder Documentation](#).

Baseline checked 10 August 2026. Verify disease-specific rules, data releases, service status, geographic vintages, denominators, and privacy policies at point of use. The case and data are synthetic.